A Topic Proposal by
Panganiban, Kyle Cedric R. | Samson, Janmikael A. | Ybut, Gian Paolo A.

Prayer

Saint John Baptist de La Salle,
Pray for us



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Web-Based Road Damage Detection and Mapping System

Using **YOLOv8** and **GPS Synchronization**

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PROJECT CONTEXT

- Web-Based Road Damage Detection and Mapping System
- Manual road inspection is time-consuming and labor-intensive
- Traditional surveys may produce inconsistent reports
- Road damages affect transportation safety and efficiency
- AI and computer vision can automate road monitoring
- YOLOv8 enables fast and accurate damage detection
- GPS synchronization improves location tracking of road damage
- The study proposes a centralized web-based monitoring platform

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Statement of the Problem

- Manual inspections are inefficient for large road networks
- Existing systems lack full GPS and mapping integration
- Some systems rely only on static image uploads
- Mobile-only systems experience hardware limitations
- Philippine road conditions affect detection reliability
- GPS synchronization inconsistencies reduce mapping accuracy
- Need for a scalable web-based road monitoring solution

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General Objective

Develop a web application that detects and maps potholes from uploaded road survey videos using YOLOv8 and GPS data

Specific Objectives

- Create a web interface for uploading videos and GPS files
- Train a YOLOv8 pothole detection model
- Detect potholes frame-by-frame using video analysis
- Synchronize detections with GPS coordinates
- Visualize potholes using interactive maps
- Evaluate detection accuracy and system usability

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Significance of the Study

- Assists road maintenance personnel in monitoring damages
- Helps LGUs improve maintenance planning and budgeting
- Supports engineers in infrastructure assessment
- Improves organization of road condition records
- Contributes to safer transportation conditions
- Serves as reference for researchers and developers
- Demonstrates practical AI and geospatial applications

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Scope

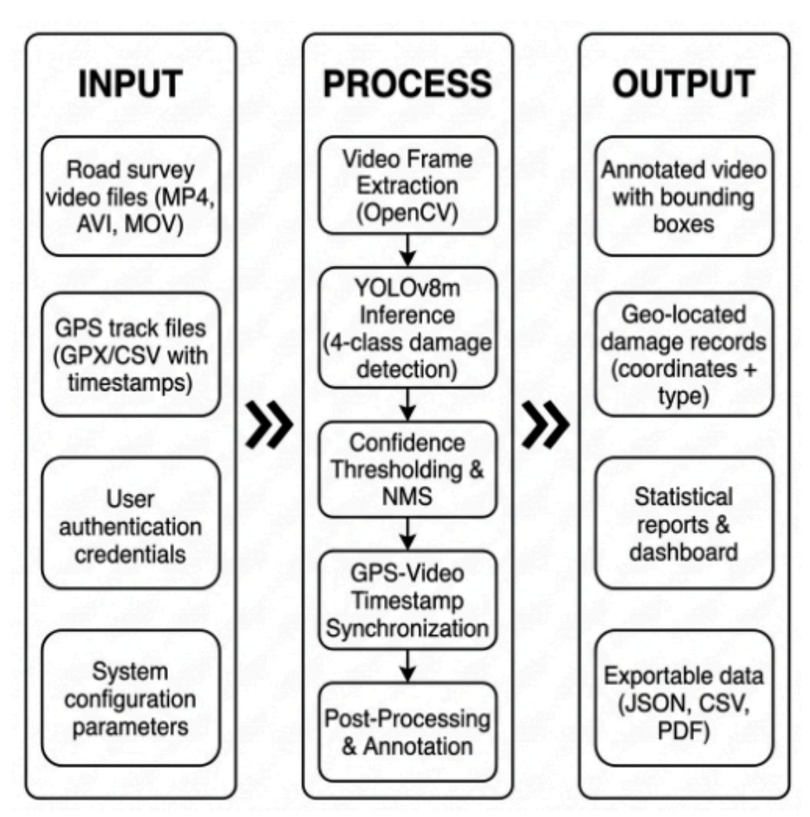
- Detects potholes, longitudinal, transverse, and alligator cracks
- Processes uploaded road survey videos using YOLOv8 frame analysis
- Synchronizes detections with GPX or CSV GPS track files
- Displays mapped detections using React-Leaflet and OpenStreetMap

Limitations

- Does not support real-time roadside mobile-device detection
- Detection accuracy affected by lighting, weather, and video quality
- GPS inaccuracies and timestamp inconsistencies may affect mapping precision

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Methodology / System Process



- Use YOLOv8 for pothole object detection
- Train the model using RDD2022 and local datasets
- Upload dashcam videos and GPX/CSV GPS files
- Process videos frame-by-frame on the server
- Extract timestamps of pothole detections
- Synchronize timestamps with GPS track data
- Display potholes on OpenStreetMap using React-Leaflet

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Algorithms Used

Algorithms and Their General Tasks

- **YOLOv8m** – Detects and classifies road damage from frames
- **CSPDarknet** – Extracts multi-scale image features efficiently
- **FPN (Feature Pyramid Network)** – Combines high and low-level feature maps
- **PAN (Path Aggregation Network)** – Strengthens feature propagation across network layers
- **Anchor-Free Detection Algorithm** – Predicts object locations without anchor boxes
- **Non-Maximum Suppression (NMS)** – Removes overlapping duplicate object detections
- **GPS Timestamp Synchronization Algorithm** – Matches detections with GPS coordinate timestamps